

At the crossroad of lifespan, calorie restriction, chromatin and disease: meeting on sirtuins.

Longevity, lifespan, cancer, cellular transformation, energy, calorie restriction, diabetes--what can tie together such a diversity of hot topics in biomedical research? Emerging findings suggest that the answer lies in understanding the functions of the recently discovered family of proteins known as Sirtuins. Barcelona hosted the first scientific meeting completely focused on these evolutionary conserved protein deacetylases, bringing together experts in the biochemistry to cellular biology, mice models, drug targeting and pathophysiology of these molecules. Their work, summarized here, establishes the Sirtuins as major players in cellular homeostasis and human diseases that act through a whole range of biochemical substrates and physiological processes. Undoubtedly, this is an increasingly expanding field that it is here to stay and growth.